IR Homework #1

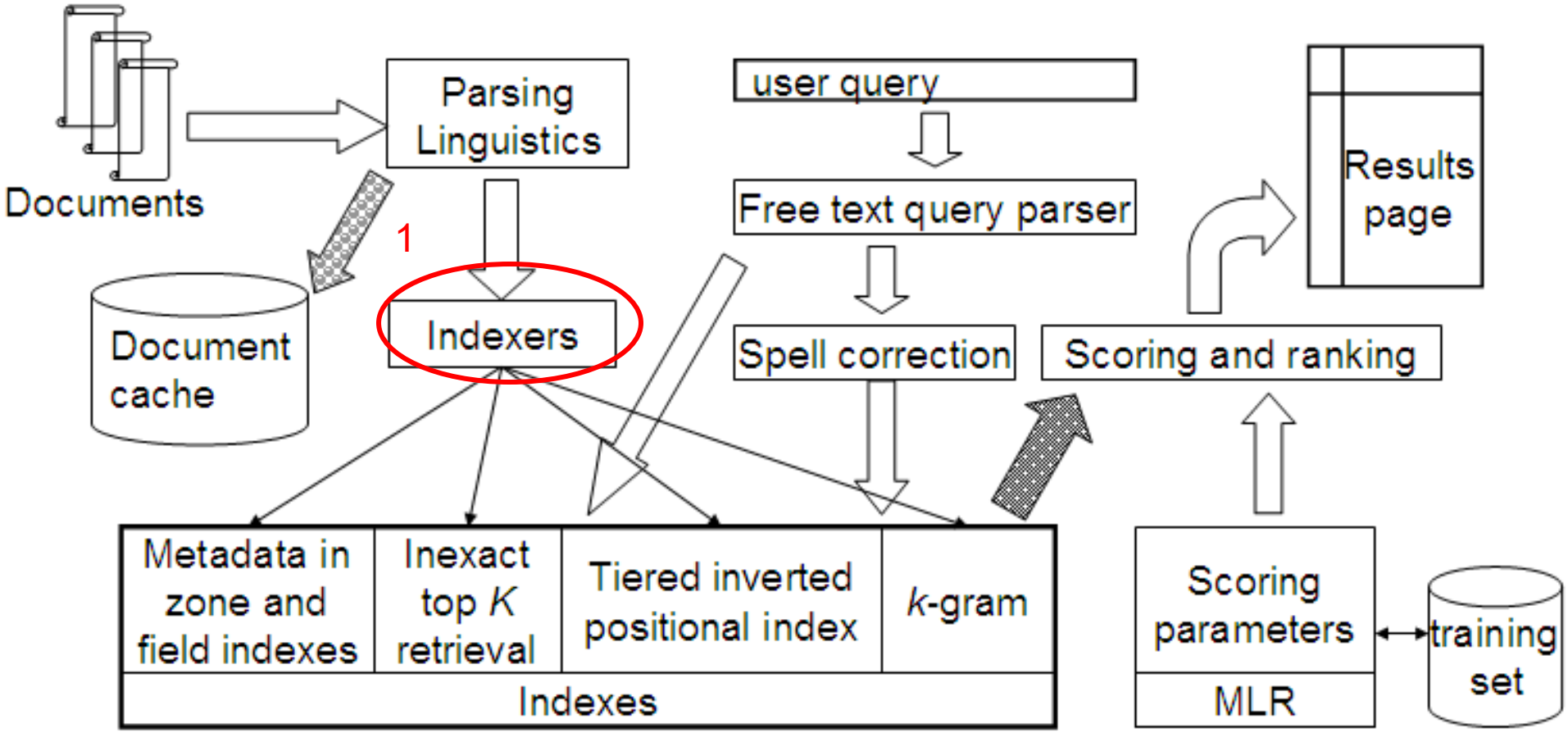
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Programming Exercise #1: Document Indexing

- Goal: to build an *inverted index* for a text collection
- Input: a set of text documents
- Output: inverted index
- Implementation:
  - either calling open source library functions (or APIs)
  - or writing your code from scratch in any programming language

The Major Task to Do



Input: a Sample of the Test Collection

- A sample from the ClueWeb09 test collection
  - File format: WARC (Web ARChive file format)
    - http://www.digitalpreservation.gov/formats/fdd/fdd000236.shtml
    - Sample Files: http://www.lemurproject.org/clueweb09/sampleFiles.php
  - Each file contains about 40,000 Web pages, in 1GB
  - Each team will be randomly allocated different files by the TA (later)

Web Test Collections

- The ClueWeb09 dataset
  - http://lemurproject.org/clueweb09/
    - 1,040,809,705 Web pages in 10 languages, during Jan.-Feb. 2009
    - 5TB, compressed (25TB, uncompressed)
    - Distributed as two 3TB hard disks for the full dataset
- The ClueWeb12 dataset
  - A successor to the ClueWeb09 dataset
  - http://lemurproject.org/clueweb12/
  - 733,019,372 English Web pages, in Feb.-May 2012
  - 5.5TB, compressed (27.3TB, uncompressed)
  - Distributed as one 8TB hard disk for the full dataset

Web Test Collections

- The ClueWeb22 dataset
  - The newest dataset: https://www.lemurproject.org/clueweb22.php/
  - 10 billion Web pages, organized into three subsets: B (200M), A (2B), L (10B)
    - Collected during the first half of 2022 (in Bing)
    - 1* 8TB disk (B), or on Azure
  - A. Overwijk, C. Xiong, and J. Callan. ClueWeb22: 10 billion web documents with rich information. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, ACM, 2022.

Other Test Collections

- Datasets previously used in TREC Web, Terabytes, and Blog tracks:
  - Currently managed by Univ. of Glasgow at: http://ir.dcs.gla.ac.uk/test_collections/
  - WT2g, WT10g, .GOV, .GOV2, Blogs06, Blogs08
- Reuters Corpora (RCV1, RCV2, TRC2)
  - http://trec.nist.gov/data/reuters/reuters.html
  - RCV1: ~810,000 English news stories from 1996/08/20 to 1997/08/19 (2.5GB uncompressed)
  - RCV2: 487,000 multilingual news in 13 languages
  - TRC2: 1,800,370 news from 2008/01/01 to 2009/02/28 (2.87GB)

Older Test Collections

- Reuters-21578:
  - 21,578 news articles in 1987 (28.0MB uncompressed)
  - Publicly available at: http://www.daviddlewis.com/resources/testcollections/reuters21578/
- Test collections held at University of Glasgow:
  - Publicly available at: http://ir.dcs.gla.ac.uk/resources/test_collections/
  - LISA, NPL, CACM, CISI, Cranfield, Time, Medline, ADI
  - Ex: The Time Collection: 423 articles from Time magazine (1.5MB)

Output: Inverted Index

- Output format: Using the standard positional index (as in Chaps. 1 & 2):
  - (1) Dictionary file: a sorted list of vocabularies (in separate lines)
  - (2) Postings list: for each term, a list of occurrences in the original text
    - Format: (as in Fig. 2.11, Sec. 2.4.2, p.38)
      $term_i$, df_i:
      <doc1, tf_{i1}: <pos1, pos2, … >;
      doc2, tf_{i2}: <pos1, pos2, …>;
      …>
    - E.g.
      to, 993427:
      <1, 6: <7, 18, 33, 72, 86, 231>;
      2, 5: <1, 17, 74, 222, 255>; … >

df_i: *document frequency* of $term_i$

tf_{ij}: *term frequency* of $term_i$ in doc_j

Design Issues

- pos means the token positions in the body of documents
  - This facilitates easier implementation of some functions, e.g., phrase query, or proximity search
- You can design your own internal index formats, as long as necessary information are included
  - Dictionary: terms t_i, document frequency df_i
  - Postings: a list of (docID, term frequency tf_{ij}, loc) for each term t_i
- Preprocessing should be handled with care
  - Input formats of data collections
  - Please indicate how you will handle digits, hyphens, punctuation marks, … in your tokenizer

Optional Functionality

- Efficiency issues
  - A separate data structure (e.g. trie) can be used to store the vocabularies and postings in your indexer
  - Skip pointers (to be used in query processing)
- Language issues in tokenization
  - Case folding
  - Stopword removal
  - Lemmatization/Stemming
  - Able to be turned on/off by a parameter trigger

Some Open Source Indexing Tools

- Apache Lucene (in Java)
  - Lucene Core (in Java)
  - PyLucene (Python binding)
  - Popular library on which search engines are built: Apache Solr, ElasticSearch
- The Lemur Project:
  - by CMU/Umass (in C++)
  - Lucindri (in Java)
- Terrier IR Platform
  - by U. Glasgow (in Java)
  - PyTerrier (Python binding)
- …

Submission

- Your submission *should* include
  - The source codes
    - For calling open source libraries, please indicate which APIs or library functions are needed and instructions on where to download, how to install, configure or compile
  - A one-page documentation including
    - **Major features**: ex: optional features, high efficiency, low storage requirement, multiple input formats, huge corpus, …
    - **Major difficulties encountered**
    - **Instructions for installation/compilation/ configuration/execution environment** (ex: Java Runtime, special compilers, …)
    - **Team members list:** The names and the responsible parts of each individual member *should* be clearly identified
- Due: two weeks (Oct. 21, 2026)

Submission Instructions

- Programs and related electronic files in your homework must be submitted directly on the submission site:
  - **Submission site:** iSchool+
  - **Preparing your submission file**: as ***one single compressed file***
    - Name your file according to your ID such as <ID>_HW1.zip
    - Remember to specify the names and student IDs of your team members in the documentation
- NOTE:
  - You have to register of your team members to the TA first, before we can allocate different files as your test input
  - Instructions on where to obtain your copy of the partial dataset will soon be available from the TA
- For more information on homework submission issues:
  - Please contact with the TA (Ms.Hsu & Ms.Liao, @ R1424, Technology Building)

Evaluation

- *Minimum requirement*: index correctness for your test data
  - Using (your allocated copy of) the ClueWeb09 Test Collection as input, the inverted index generated by your program will be checked
  - Optional features will be considered as bonus
- You might be required to demo if the program submitted was unable to compile/run by the TA

Any Questions or Comments?